



A PROJECT PROPOSAL

(Submitted for the degree of Bachelor of Science in Information Technology)

CLOUD-BASED LOGISTICS & FLEET MANAGEMENT SYSTEM

(Transport Management System)

(Digitizing Manual Ledger & "Hisab Kitab" for Trucking Operations)

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Project Title: Cloud-Based Logistics & Fleet Management System (Transport Management System)

Developer: Khuram Shehzad – IT Department, IUB

Tech Stack: TALL Stack (Tailwind, Alpine.js, Laravel, Livewire) + Filament PHP

1. Executive Summary

This project aims to digitize the manual bookkeeping (Hisab Kitab) of a transport business managing 35-40 trucks. By replacing the traditional register-and-pen method with a robust web-based solution, the business will achieve real-time tracking of earnings, driver expenses, and vehicle maintenance, ensuring zero data loss and improved financial transparency.

2. Problem Statement

The current manual system faces several critical challenges:

- **Data Vulnerability:** Physical registers are prone to damage, loss, or unauthorized alterations.
- **Calculation Errors:** Manual "Hisab" often leads to accounting errors in driver salaries and fuel expenses.
- **Lack of Reporting:** It is difficult to calculate the monthly profit/loss or performance of a specific vehicle quickly.
- **Operational Delay:** Booking and checking vehicle availability requires searching through pages of paper.

3. Proposed Solution

The proposed solution is a centralized **Fleet & Financial Management System** built using the **TALL Stack**.

- **Filament PHP** will be used to provide a high-performance administrative panel.
- The system will handle everything from vehicle booking to complex financial transactions and driver payouts.

4. Key Features (Functional Requirements)

The system will be divided into the following modules:

A. Fleet Management

- **Vehicle Profiles:** Digital records for 40+ trucks (Model, Plate No, Fitness/Insurance dates).
- **Driver Management:** Records of driver licenses, contact info, and assigned vehicles.

B. Booking & Trip Management

- **Consignment Entry:** Recording what is being shipped, the destination, and the agreed rate.
- **Trip Logs:** Auto-calculating the distance and duration of trips.

C. Financial Management (The "Hisab Kitab" Core)

- **Expense Tracking:** Recording fuel costs, tolls, repairs, and police "kharcha."
- **Revenue Management:** Tracking payments received from clients vs. pending balances.
- **Driver Salaries:** Automated calculation based on trips completed or fixed monthly rates.

D. Reporting & Analytics

- **Profit/Loss Reports:** Daily, weekly, and monthly financial summaries.
- **Vehicle Performance:** Identifying which trucks are costing more in repairs than they earn.

5. Technical Stack

- **Backend:** Laravel 11 (PHP)
- **Admin Panel:** Filament PHP (v4) for rapid, secure dashboarding.
- **Frontend:** Livewire & Alpine.js for a reactive, "single-page app" feel without page reloads.
- **Styling:** Tailwind CSS for a modern, mobile-responsive design.
- **Database:** MySQL for relational data integrity.

6. Development Timeline (7th & 8th Semester)

1. **Phase 1 (Month 1):** Requirement Analysis & Database Schema Design.
2. **Phase 2 (Month 2):** Core Module Development (Vehicles & Drivers).
3. **Phase 3 (Month 3):** Financial Module & Ledger Implementation.
4. **Phase 4 (Month 4):** Reporting Engine & Final Testing.
5. **Phase 5 (Month 5):** Deployment & Client Training.

7. Expected Benefits

- **Accuracy:** 100% accurate financial calculations.
- **Accessibility:** Management can check reports from their mobile phones via the web app.
- **Scalability:** The system can easily handle 100+ trucks if the business grows.